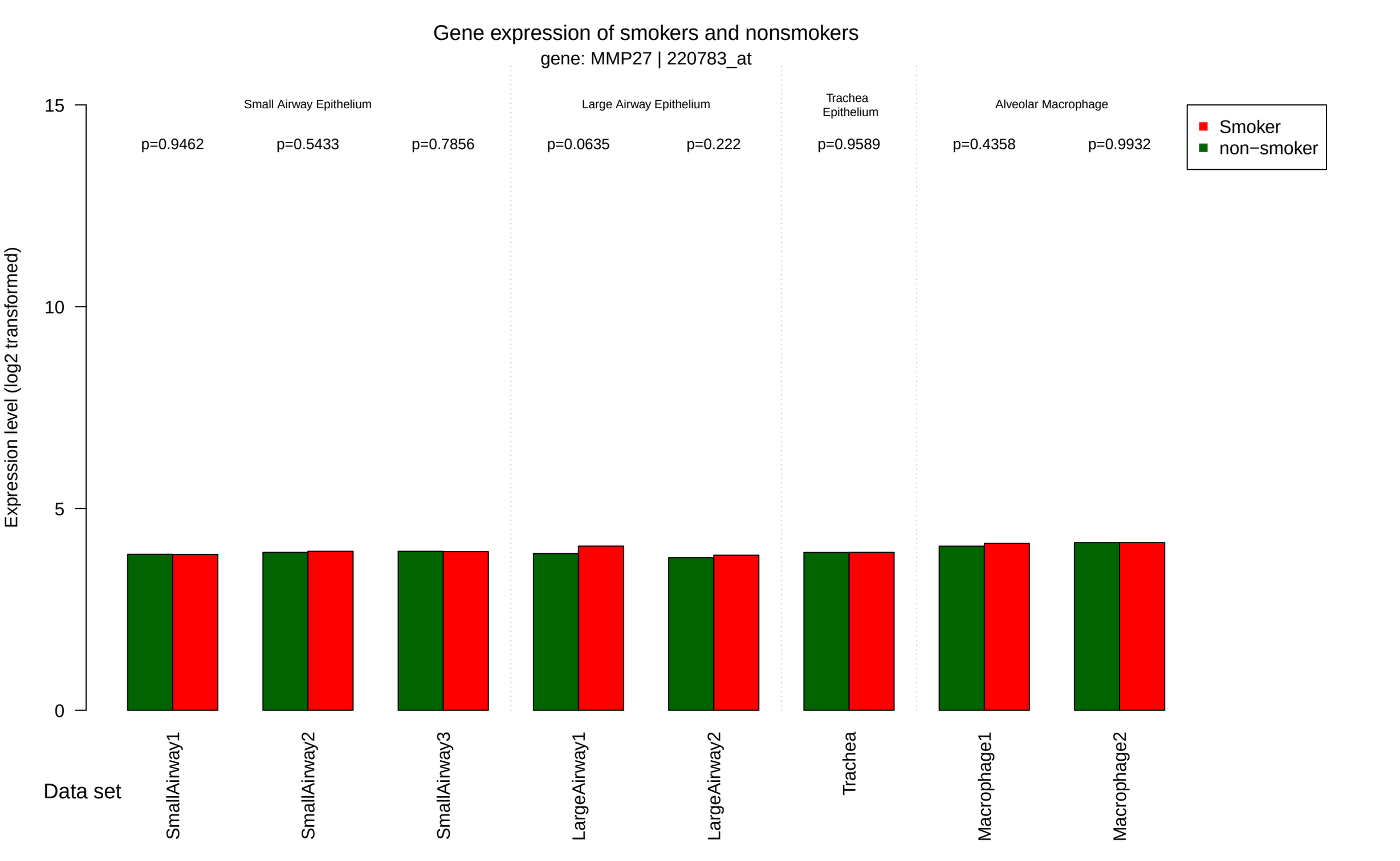
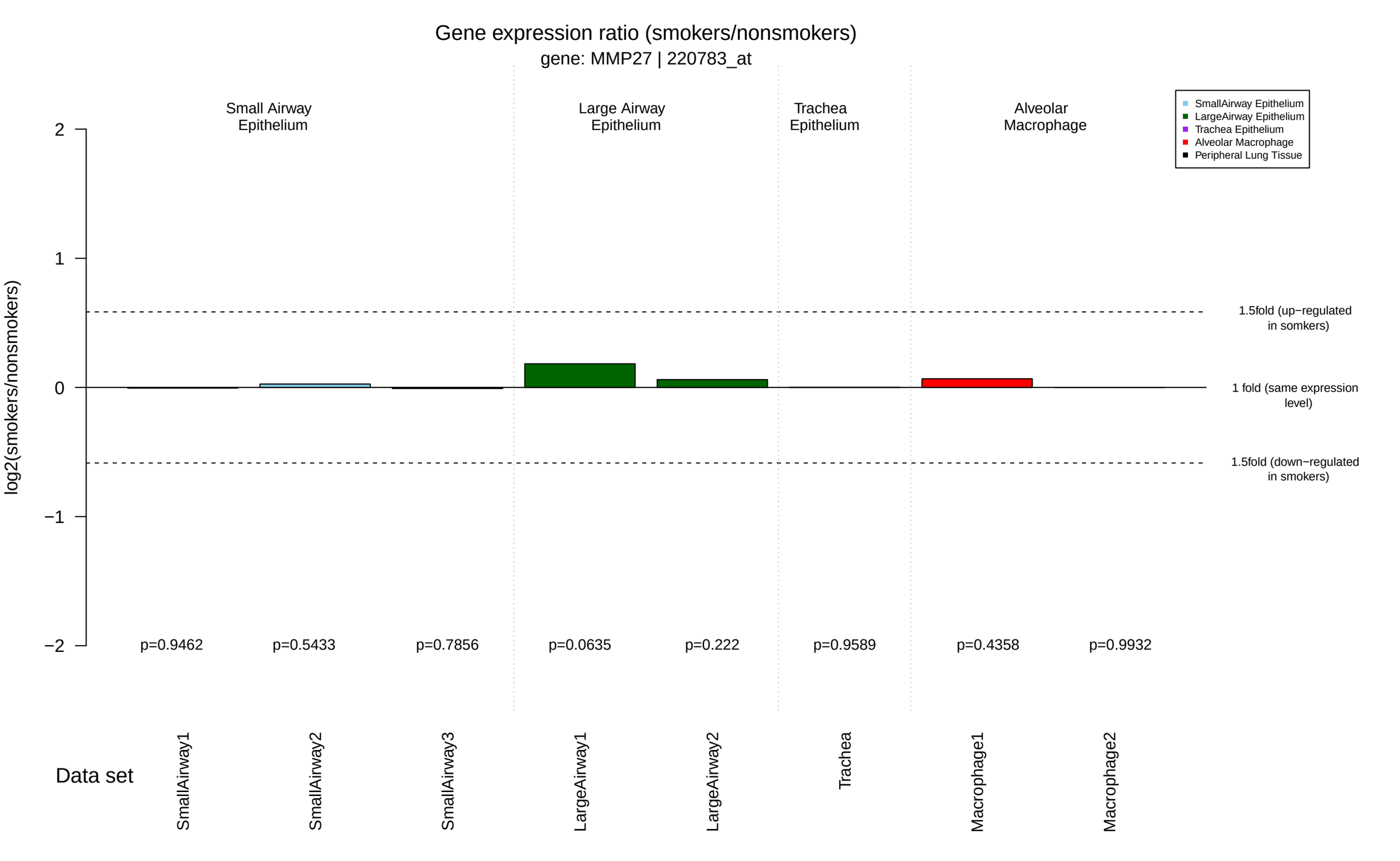
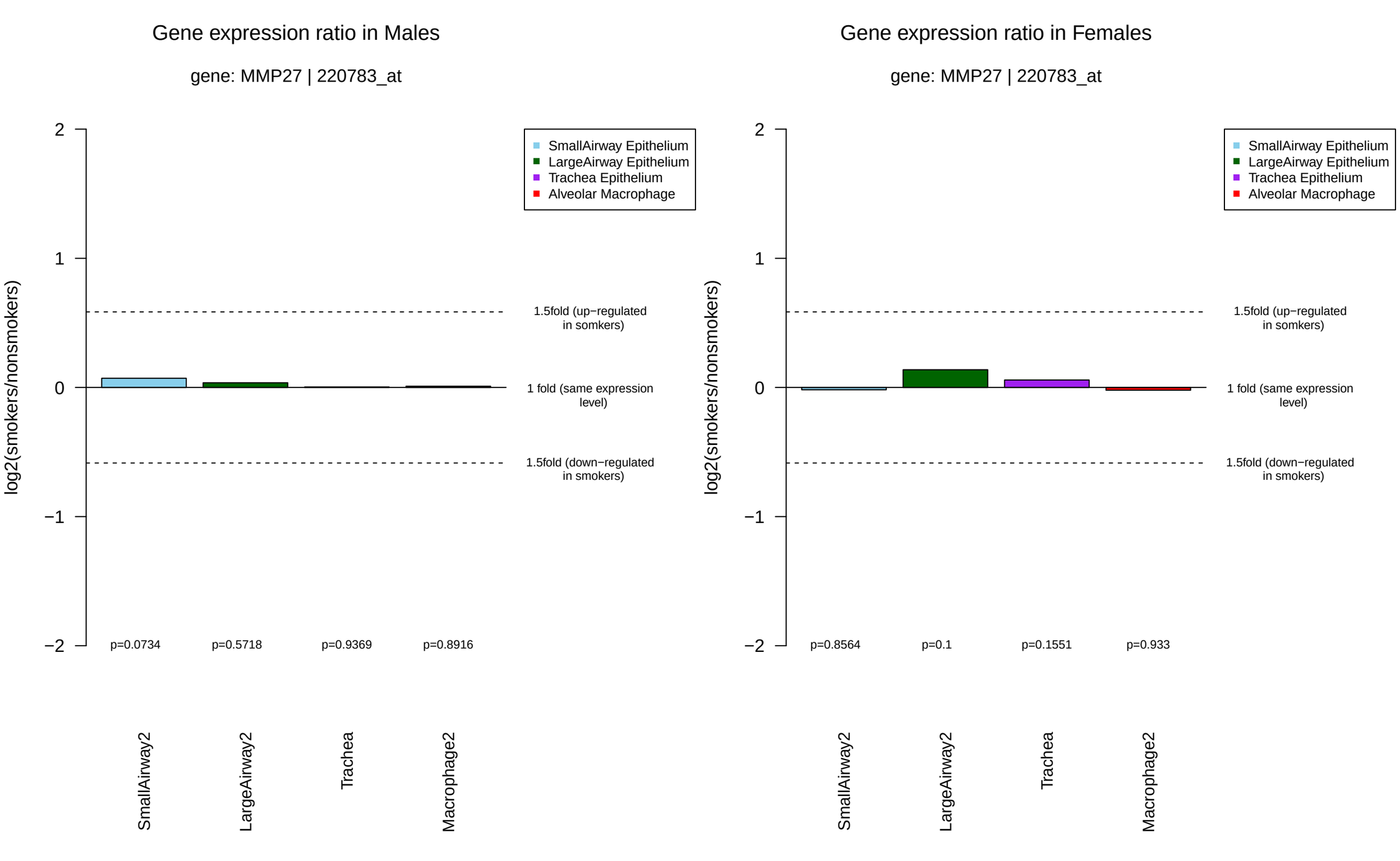
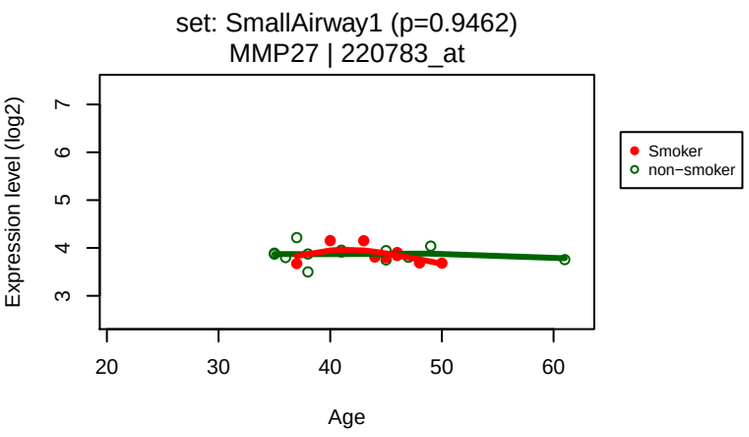
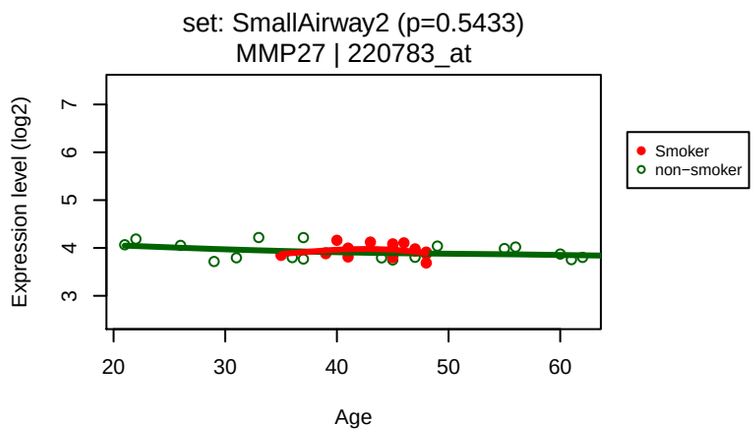
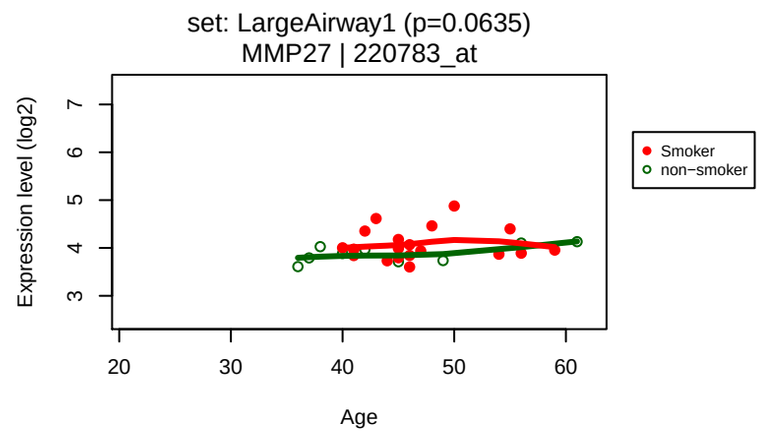

Gene expression of smokers and nonsmokers
gene: MMP27 | 220783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9462
p=0.5433
p=0.7856
p=0.0635
p=0.222
p=0.9589
p=0.4358
p=0.9932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

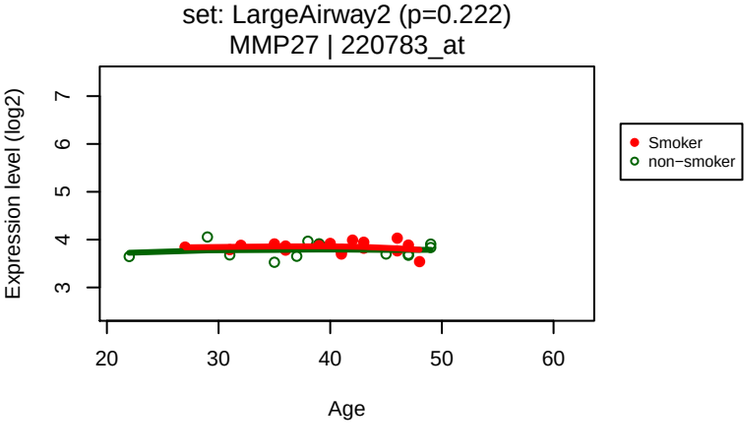

Gene expression ratio (smokers/nonsmokers)
gene: MMP27 | 220783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9462
p=0.5433
p=0.7856
p=0.0635
p=0.222
p=0.9589
p=0.4358
p=0.9932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

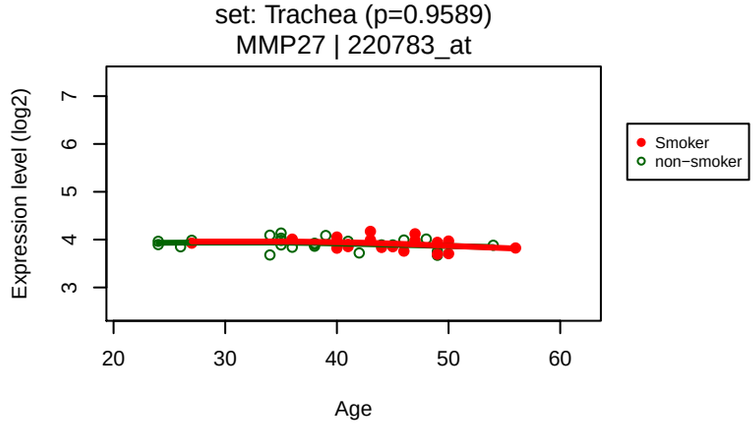

Gene expression ratio in Males
gene: MMP27 | 220783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0734
p=0.5718
p=0.9369
p=0.8916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP27 | 220783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8564
p=0.1
p=0.1551
p=0.933
SmallAirway2
LargeAirway2
Trachea
Macrophage2

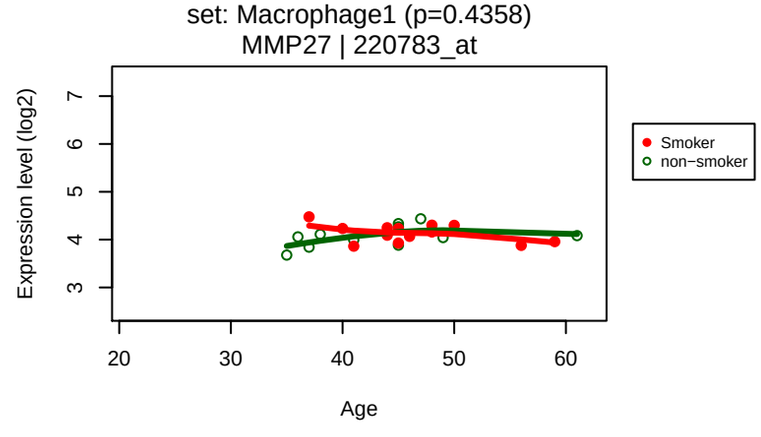

set: SmallAirway1 (p=0.9462)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker

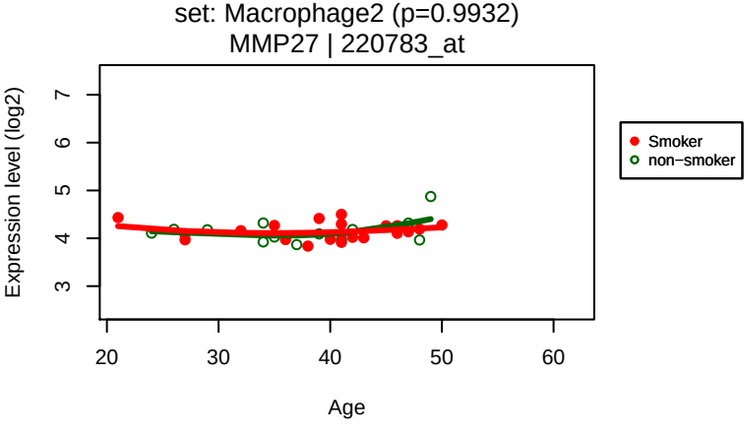

set: SmallAirway2 (p=0.5433)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker

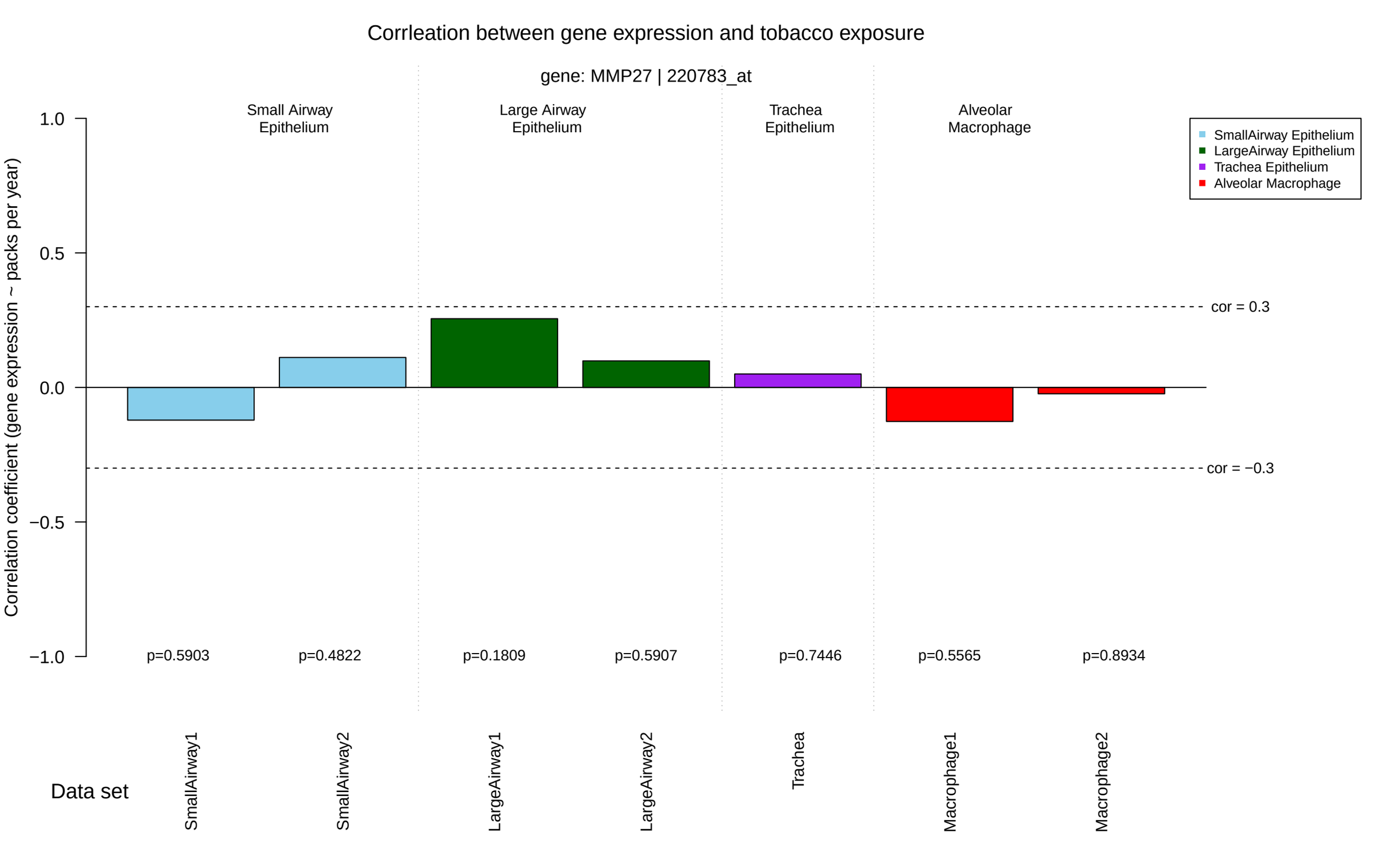

set: LargeAirway1 (p=0.0635)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.222)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9589)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4358)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9932)
MMP27 | 220783_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP27 | 220783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5903
p=0.4822
p=0.1809
p=0.5907
p=0.7446
p=0.5565
p=0.8934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2